

SAIRAJ PANKAR

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SUMMARY

AI/ML Software Engineer with 1+ years of hands-on experience building and deploying machine learning, computer vision, and real-time AI systems for industrial applications. Strong in Python, PyTorch, YOLO, Scikit-learn, FastAPI, CUDA, and Linux, with experience spanning model development, GPU-accelerated inference, distributed software systems, and automated deployment. Optimized vision pipelines handling 70+ concurrent IP and thermal camera streams, reducing CPU utilization from 80%+ to 35-40% through GPU acceleration and resource optimization.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java
- **AI/ML & Applied AI:** PyTorch, Scikit-learn, YOLO, OpenCV, LightGBM, XGBoost, CatBoost, SentenceTransformers, Pandas, NumPy, Embeddings, Semantic Search, Pose Estimation
- **Backend & Systems:** FastAPI, Flask, REST APIs, SQLAlchemy, PySide6 / PyQt, Multithreading, Distributed Systems, Linux
- **Databases:** PostgreSQL, MySQL, MongoDB, Pinecone
- **Cloud & DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD, Kubernetes, Tekton, Gunicorn, PyInstaller, Debian Packaging
- **Performance & Edge:** CUDA, GPU-Accelerated Inference, RTSP Streaming, Real-Time Video Processing, Linux SBC Deployment

EDUCATION

Bachelor of Engineering (B.E.) in Electronics and Computer Science — Shree L.R. Tiwari College of Engineering Jul 2021 – Jul 2025

CGPA: 8.73 / 10

EXPERIENCE

Software Engineer (R&D) — Blujay Robotics Pvt. Ltd.

Jun 2025 – Present | Mumbai, India

- Built a **thermal-camera fire classification pipeline** trained on **42K+ fire/non-fire samples**, engineering **63 temperature, spatial, intensity-gradient, distance, and temporal-flicker features** with a Scikit-learn **Gradient Boosting** pipeline for real-time fire detection in an AI-enabled industrial fire-response system.
- Developed and deployed a **lightweight YOLO-based human detection and pose-estimation pipeline** for live RTSP streams, using **17-keypoint pose inference** to classify **standing, sitting, sleeping, and person-missing states** while maintaining **<15% GPU utilization** for real-time deployment.
- Architected a **distributed Python software system across 2 Linux SBC controllers**, separating perception, navigation, thermal imaging, device-control, and recovery processes while coordinating real-time inter-controller communication.
- Automated application builds and Linux packaging using **GitHub Actions, Python, PyInstaller, and Debian tooling**, eliminating a **~2-hour manual release workflow** and enabling authorized users to retrieve deployment-ready software without engineer-assisted builds.
- Developed an **AES-256-GCM application-layer security module** to protect RTSP credentials and sensitive model/application paths in deployed systems, using externalized key management to prevent direct exposure at customer sites.

Software Engineering Intern — Blujay Robotics Pvt. Ltd.

Jan 2025 – Apr 2025 | Mumbai, India

- Contributed to the **ground-up development of Airavat**, GPU-accelerating real-time vision workloads using **PyTorch and CUDA across 50+ IP and 21 thermal camera streams**, reducing CPU utilization from **80%+ to 35-40%** while utilizing approximately **20% GPU**.
- Engineered an **on-demand RTSP streaming pipeline** that initializes camera streams in the backend and renders frames only when requested, automatically releasing cached resources when streams are closed to reduce unnecessary memory and compute utilization.
- Re-engineered Airavat's **SCADA visualization and mapping system from Matplotlib to PyQt**, supporting site-level visualization of cameras and robotic water monitors while eliminating application crashes and memory spikes from the previous implementation.

PROJECTS

Home Loan Default Risk Prediction

- Developed a **stacked ensemble for imbalanced loan-default prediction** using LightGBM, XGBoost, and CatBoost, achieving **0.88 ROC-AUC** on Home Credit default-risk data.
- Engineered domain-specific financial, external-risk, missingness, and behavioral features, implementing **leakage-safe categorical encoding, stratified 5-fold cross-validation, and out-of-fold stacking** for robust model training and evaluation.
- Integrated the trained ensemble into an end-to-end application using **FastAPI, PostgreSQL, Docker Compose, and GitHub Actions**, exposing model inference through REST APIs with automated testing and container builds.

RetailDialogue — Conversational Semantic Search

- Engineered a **semantic product-discovery pipeline** using SentenceTransformers, Pinecone, FastAPI, and Dialogflow, transforming natural-language queries into embeddings for vector similarity search and retrieval of **semantically relevant products**.
- Implemented multi-turn conversational commerce workflows for product search, cart management, order placement, and order tracking using **Dialogflow intent detection, FastAPI webhook fulfillment, and session-based state management**.

IBM DevOps Capstone — Cloud-Native Account Microservice

- Implemented a **Flask/PostgreSQL REST microservice** using SQLAlchemy and test-driven development (TDD), covering CRUD operations, validation, structured error handling, health checks, and automated model/API testing.
- Containerized the service with **Docker/Gunicorn** and implemented delivery workflows using GitHub Actions, Kubernetes, and Tekton, including automated linting/testing and a **3-replica Kubernetes deployment** backed by PostgreSQL.